I'm in my Tesla on autopilot going 40 mph towards a fake Wile E Coyote Roadrunner painted wall.

0:06

Please stop.

0:07

Please stop.

0:08

Holy...

0:08

- And I'm doing this to see if Tesla's autopilot can be tricked.

0:11

Because it famously only relies on simple cameras to navigate the world as opposed to much more

0:17

expensive tech.

0:18

In fact, we're out here today to run a bunch of tests in extreme conditions to see how those

0:23

simple cameras on a Tesla stack up against other fancy cars that use more advanced technology.

0:28

- Wow!

0:29

- And the story of how I ended up as a crash test dummy in my own car...

0:32

- Oh gosh!

0:33

(screams)

0:34

- started 6 months back at Disneyland as I was sneaking onto the famously pitch black ride called

0:39

Space Mountain.

0:40

And while I might look a little plus size in that jacket, as you can see from this infrared camera

0:45

shot, that's no ordinary jacket because it's hiding some of that fancy car technology.

0:51

- Yeah baby!

0:55

- 43 million people ride Space Mountain every year, but unlike a famous ride like Colossus, Space

1:01

Mountain is in near total darkness, meaning no one knows what the ride actually looks like.

1:08

Which, as far as I'm concerned, needed to be fixed with a special undercover operation.

1:12

- True story, I have been wanting to make this video for over 20 years, way before I had a YouTube

1:17

channel because I'll come to Disneyland and just be like, someday the technology will be there so

1:22

we could map out what Space Mountain actually looks like.

1:25

So this is a very big moment for me.

1:27

- It's common knowledge every good undercover operation worth its salt needs a 3-step plan.

1:32

For me, step one would just be getting into Disneyland with my tech through their famously

1:36

tight security.

1:37

And if I managed to do that, step 2 would be making it through the park, avoiding any security

1:41

guards on my way to Space Mountain.

1:43

And finally for step 3, I would need to sneak onto the ride itself while avoiding any suspicion to

1:48

make my childhood dream come true.

1:50

And if all this seems like overkill, let's just say I have a bit of history

1:53

- First row, please put your phone away!

1:54

- getting kicked out of amusement parks in the name of science.

1:57

- I follow you, man, why are you doing this to me?

2:00

- For step one, I took an oversized jacket and made a little pocket inside where I could hide my

2:04

secret tech inside it.

2:05

- It's like the slight limp shuffle walk that really sells it, I think.

2:09

- Because they make you walk through metal detectors, I disguised and temporarily placed my

2:13

device with a bunch of gear in my camera bag and just had to hope it wouldn't get discovered.

2:18

- Can I walk through?

2:19

- We'll just wait until your bag is fully screened in case you have items that are not supposed to be there.

2:24

- And even though this moment of truth was terrifying, I knew I just needed to play it cool.

2:30

Like that.

2:30

- Excuse me can you not record?

2:32

- And the security was in fact extremely tight.

2:35

But our alibi was tighter.

2:37

- All right, we're in 1 down, 2 to go.

2:40

- Once inside, I immediately made my way to a bathroom to properly suit up.

2:43

Now I just had to make it all the way over to Space Mountain without arousing the suspicion of

2:47

any of the park's security.

2:48

Which is a lot harder than you might think because it's a well kept secret Disney employs a bunch of

2:53

security officers dressed up in plain clothes as normal parkgoers.

2:56

So I could trust no one.

2:58

So in an attempt to blend in and look like a normal park goer, I took a corn dog break.

3:03

- We're doing this for the integrity of the mission.

3:08

And not because they're incredibly delicious.

3:10

- Back on track after a close call or two, I had made it inside the building.
3:14
- All right, 2 down now for the hard part.
3:16
- Now that we're inside, it's probably a good time to explain why my jacket looks like this when you
3:20
view it with an infrared camera.
3:22
Because that is a really tiny portable LiDAR scanner.
3:25
So it's shooting out 640,000 laser pulses per second to fully map the surroundings.
3:30
Now as for the legality of this, if you check the back of your ticket, it says you have to obey all
3:35
the park rules.
3:36
And if you check their website for the rules on forbidden items, it says you can't bring in shoes
3:41
with wheels or stun guns or cremated remains.
3:43
Ew.
3:44
But you'll notice the forbidden items list makes no mention of chest-mounted LiDAR scanners.
3:49
So technically, I'm not breaking any laws by precisely mapping out this entire ride down to the
3:54
nearest half inch.
3:56
- Don't blow this Rober!
3:57
We got this!
3:58
- After sneaking my way past the last of the ride security, I was finally in place.
4:02
And this is probably a good time to point out that the final boss here in this control room watching
4:06
all the infrared see-in-the-dark cameras was the one I was most afraid would shut down the whole operation.
4:11

Because when we tested this out to see how inconspicuous the LiDAR would be, as you can see here,
4:16
not very,
4:17
as it lights up the whole place like a Christmas Tree.
4:19
By the way, you should know LiDAR stands for light detection and ranging, whereas radar is radio
4:25
detection and ranging.
4:26
Basically, it's like having a bunch of laser pointers inside the shell shooting all around the room.
4:31
And each time you fire one of them, you then see how long it takes for you to see it hit the object
4:35
it's pointed at.
4:36
And since we know the speed of light, if we measure that time, it will tell us how far away
4:40
the object is.
4:41
For example, it will take twice twice as long to detect the laser point on an object 20 feet away,
4:45
as opposed to an object that's 10 feet away.
4:47
But how could the sensor possibly keep track of 640,000 laser pulses firing all out at once?
4:53
Well, it turns out, amazingly, they actually happen one after another, so the sensor only has
4:58
to keep track of one pulse at a time.
5:00
And this works because the speed of light is so fast in fact, if you slow time way down,
5:04
so the time it takes a laser to hit an object is four "Slo-mo" seconds, then you can wait 30
5:09
minutes on that "Slow-mo" time scale before sending the next pulse.
5:13

So it's actually really easy for the sensor to know exactly which laser pulse it's measuring,

5:19

even if that happens 640,000 times in a single second.

5:22

And with that, it was time for the execution phase of Secret Undercover Operation Mousetrap.

5:27

- Let's go!

5:30

(screaming)

5:36

- It felt so good to cross off a 30 year old bucket list item.

5:40

- (screaming)

5:43

- But just as I feared, as our exact cart pulled back into the station, they stopped the ride and

5:48

announced it was being locked down.

5:50

- All flights have been put in a holding pattern.

5:53

- Which meant I was about to have an unscheduled appointment with a bunch of dudes with muscles

5:56

much bigger than mine.

5:57

- Your flight will continue in...

5:59

- But it turns out it was an unrelated power issue which meant this was a mission accomplished.

6:04

- That's like a dream come true.

6:05

Before I even had a YouTube channel I wanted to do that.

6:08

- And the only critical oversight of the whole operation was forgetting they took those dumb key

6:13

chain photos at the end of the ride, leaving behind some very incriminating evidence at the scene.

6:18

- I forgot to put it away!

6:19

- That's so awesome!

6:20

- That's Harrison, by the way.

6:21

And he started the company that makes these super small portable LiDARs.

6:25

After wrapping out a few more rides like Haunted Mansion

6:27

- Let's go!

6:29

- and continuing my painstaking efforts to blend in, we headed home so Harrison could process all the data.

6:34

And a few days later he did not disappoint.

6:37

- This is from the LiDAR you were wearing on your chest.

6:39

- Get out of here!

6:40

- All of this is totally in the dark.

6:41

- Now, what's really cool is we could visualize what was recorded as the ride progresses.

6:45

And so as you can see here, you get a sense of what the LiDAR was seeing in real time as the ride

6:49

progressed on.

6:50

But what's even cooler though is we can now use that data to make an actual tabletop 3D model of

6:55

Space Mountain with the help of our army of 3D printers.

6:58

Which meant for the first time ever, this question mark's days were numbered.

7:04

- (extinguishes)

7:06

Got it!

7:07

- So while all the 3D printers were hard at work, we also reviewed the data we captured from the

7:11

haunted mansion ride, where we made two super interesting discoveries.

7:14

The first is that the initial room they take you in is really just a fancy elevator so they can get

7:19
you underground to transport you to the real ride, which actually isn't in this house at all.
7:24
Because when you go down this hallway, you're walking under this hill you see here.
7:27
So by the time you load onto the ride, you're totally outside the park.
7:31
But if that's true, there should be some evidence for this besides our scan, and sure enough, There
7:36
is, because if you look back at this shot of the hill, you can actually see just the corner of some
7:41
kind of large building.
7:42
And if you check that alongside Google Maps, you could see this entire massive, above-ground
7:47
warehouse where the ride actually takes place that matches up perfectly with our LiDAR scan.
7:52
Which is just super clever engineering by the super talented Disney imagineers.
7:56
And the second discovery is they use a lot of super thin, nearly invisible curtains they project
8:00
on that create a lot of the illusions like these flying ghosts.
8:04
And while your eye might get fooled, as you can see here, there's no hiding from the LiDAR scan.
8:10
- And so you can imagine there are some incredible real world applications for the LiDAR technology,
8:15
Like every day when your phone scans your face to unlock itself, or when archaeologists use it to
8:20
find lost ancient cities from the sky.
8:22
Or, for example,
8:24
(car braking)
8:27
... in cars.
8:28

- And as you might guess, this car automatically saved itself from crashing because it uses a LiDAR sensor,

8:32

which as you can imagine, would have saved Wile E Coyote a lot of pain over the years.

8:39

And the reason it works so well is because it's updating that point cloud of 640,000 laser

8:43

measurements every second to the brain of this car showing exactly what's in front of it.

8:48

In fact, it works so well, as you can see here, I can maneuver all around these obstacles in real

8:54

time with all the windows blacked out, looking just at that point cloud in a VR headset.

8:59

- OK, easy bike guy.

9:00

- My Tesla on autopilot, however, only relies on simple cameras and its image processing to

9:05

navigate the world.

9:06

So to see if that tech is just too simple we came up with a six-part LiDAR versus camera head to

9:11

head, face off, culminating in a history making first Tesla-versus-cartoon-physics test.

9:16

- OK, so we're starting out with the car with the LiDAR.

9:19

First test is this kid standing in the road.

9:21

Not sure why he's doing that.

9:22

It's very unsafe, so we want to be extra careful.

9:25

OK, hit it.

9:26

- The testing speed was 40 miles an hour.

9:28

- Should have put my seat belt on!

9:29

Which meant the LiDAR would have to detect the kid and then slam on the brakes at least 60 ft in

9:33

front of him, and it turned out...

9:35

(screams)

9:36
- That's all it needed.
9:37
- (laughing)
9:39
- Now it was Tesla's turn.
9:40
- This is a terrible feeling - driving straight at a kid, but this is for science.
9:45
All right, we are up to speed.
9:47
With just simple cameras the Tesla was speeding fast.
9:50
Oh no!
9:52
- But did detect the kid.
9:54
- Oh sh...
9:55
(laughing)
9:58
- Just not in time to fully hit the brakes.
10:00
- Oh no!
10:02
He was even a CrunchLabs fan.
10:05
- If you're a Tesla owner, there is a silver lining
10:07
because we were relying on the automatic emergency braking system to stop for the car,
10:12
and because it assumes the driver's paying full attention while fully driving the car, it only
10:16
needs the brakes when it's 100% sure there's a problem in order to avoid false positives.
10:20
So the alternative is to use autopilot, and that assumes the driver isn't paying much attention and
10:25
while the downside is you get way more phantom braking and false positives
10:29
- (laughing)
10:31
That was ridiculous.

10:32
- The upside is you're less likely to be charged with vehicular manslaughter.
10:36
Because you can see here on autopilot it actually stopped in time.
10:39
So I decided to be nice and call the score 1 to 1.
10:42
And then I'd be even nicer by using the more conservative autopilot on the Tesla for all the
10:47
remaining tests.
10:48
Such as this one where we now simulate the kid dashing out from behind a parked car using some
10:53
clever engineering,
10:54
giving the cars less than a second to identify the child and stop themselves.
10:57
- Here we go, getting up to speed.
11:00
Oh, a ball came out.
11:01
I wonder why that happened.
11:03
Oh, kid!
11:03
(laughing)
11:06
Another life saved!
11:08
- Now it was the Tesla's turn on autopilot.
11:10
- 40 mph, there goes the ball.
11:13
- And impressively, it stopped with plenty of room to spare.
11:16
- Good job, Tesla!
11:17
- Which meant we were all tied up heading into round 3, the fog round.
11:21
- Optically, with my own eyes, I can no longer see that there's a kid through this fog.
11:27
The LiDAR has no issue.
11:29
This will be interesting.

11:31
OK, here we go, a wall of fog.
11:34
- So we plunged through the fog coming to a very sudden brake.
11:36
- (cheers)
11:39
- After which we saw the mannequin was not only still standing, but it was casting a really cool
11:43
long shadow, because the lasers don't pass through solid objects, just like how you can cast a long
11:48
shadow with the sun because light doesn't pass through solid objects.
11:52
- I don't have high hopes here.
11:54
But would the kid look just as cool after the Tesla test?
11:57
Oh, oh, oh! Fudge!
12:00
(laughing)
12:02
- I mean, so it does make everything look sort of cool, but that wasn't the only shock.
12:07
- I actually hit the brakes there.
12:09
- That was on autopilot.
12:12
- The cameras didn't even hit the brakes at all.
12:15
- The only thing still on is the pants.
12:18
That was a bad one.
12:19
- Now that LiDAR had taken another W
12:21
- It's time to make it rain!
12:24
- The next test would see if the cars could spot the kid under a torrential downpour made up of maybe
12:29
too much water.
12:31
- This is really interesting.
12:32

See, the Tesla can see the kid, but as soon as it starts raining, the kid is gone.
12:36
- And it was similar in the LiDAR car where you first got a clear image of the kid and the shadow,
12:41
and once we started the hose,
12:43
- oh, you see all the water going in.
12:45
LiDAR might struggle here.
12:46
OK, here we go.
12:47
- And as the wall of water started, LiDAR seemed to not slow down at all
12:51
- Oh boy!
12:52
(giggling)
12:55
- until the last possible second.
12:57
- Another W.
12:58
Now I just need the rain to stop.
13:00
You know any good rain dances?
13:01
- (rain)
13:02
I could just back up.
13:03
- Oh, that's actually a good idea.
13:04
(laughing)
13:05
- Now the LiDAR surprised me.
13:07
It was time to see if Tesla could as well.
13:09
- Oh boy.
13:11
Oh boy.
13:11
(laughing)
13:15
- And sadly It did not.

13:18
And though I did everything I could,
13:19
- Yeah, he's no longer with us.
13:21
- So we moved on to the penultimate Round 5, where we had 6 of the brightest lights that money could
13:26
buy, simulate either sunset or sunrise, or a truck in the road having its bright lights on.
13:30
Would these lights keep the cars from detecting the kid?
13:33
- Blinding trucker light test.
13:34
Let's go.
13:35
- LiDAR could immediately see much more than my eyes could when the lights turned on.
13:39
- Oh, that's a bright light.
13:41
- The question remained if it would see the kid.
13:43
- Oh, come on, LiDAR.
13:44
(laughing)
13:47
- And it did it, no problem.
13:48
- It just always waits till the last minute.
13:50
It gets me every time.
13:52
That's very bright.
13:53
Time to be blinded by the light.
13:57
- And it's true, these lights were so bright, I wouldn't have seen this kid on my own.
14:01
And at this point, I really doubted that the Tesla could either.
14:04
But when it pulled to a stop,
14:05
- OK, we stopped for the kid with the bright lights, a W for the Tesla.
14:11

- And that W meant these were the scores headed into the painted brick wall ultimate grand finale.

14:16

Scientific papers have actually Been written on this exact scenario, debating theoretically what

14:20

the Tesla would do, but we were here once and for all to silence all the debates with cold hard data.

14:26

- All right, so LiDAR is perfect and Tesla is 3 for 5, but as far as I'm concerned at this point, this

14:31

one's the only one that matters.

14:33

- Oh my gosh, I can't believe I'm gonna do this.

14:36

All right, hit it.

14:38

- Now as you can see as a human driver, while that looks sort of convincing, the image processing in

14:42

our brains is advanced enough that we pick up on minor visual inconsistencies and we wouldn't hit it.

14:47

- And as for the LiDAR, it isn't looking at what image is printed on the wall.

14:50

- So this sort of just looks like a wall, which would make this the easiest test of the day, and

14:55

as I suspected,

14:57

- Handled it with no problem.

14:58

- So the question was, would the Tesla detect and stop for this Wile E Coyote style painted wall

15:03

that was hiding yet another child staring at nothing during what might be his last minutes on Earth.

15:08

- Only one way to test that hypothesis.

15:11

I was actually supposed to go but I chickened out.

15:12

(grumbling)

15:15

The car may not know it's a wall, but I know it's a wall.

15:18
Everything in your body says, don't drive into a wall.
15:22
Oh boy!
15:24
All right, here we go.
15:24
- And so I steeled myself and accelerated the Tesla up to the 40 mph.
15:29
And as the wall crept closer and closer without moving an inch,
15:33
- Holy crap!
15:34
- The question was if the Tesla would detect it in time to step on the brakes, and it turned out...
15:39
- Holy...
15:39
(screaming)
15:48
- So I can definitively say for the first time in the history of the world, Tesla's optical camera
15:54
system would absolutely smash through a fake wall without even a slight tap on the brakes.
15:59
- (laughing)
16:00
My heart is gonna beat out of my chest!
16:05
- (lauging)
16:06
Turns out my Tesla is more Wile E Coyote and less Roadrunner.
16:10
Sorry, little fella.
16:11
Looks like we lost two arms and a head.
16:13
Not sure that's salvageable.
16:15
I think I might Uber home.
16:16
- But that's not all folks, because while LiDAR proved itself the superior car technology, could
16:21

it actually deliver on the dream of 14-year-old Mark Rober and map out Space Mountain well enough

16:27

for me to make a scaled 3D printed model?

16:29

And after 30 years of waiting, I'm happy to report for the first time ever, this is what the Space

16:34

Mountain track actually looks like.

16:36

- Whoa.

16:38

This is so cool!

16:41

Oh, there's me.

16:42

There's me; little backwards hat.

16:45

Yeah!

16:48

I recognize every single one of these turns.

16:50

You go up right there and then you come down.

16:53

Woo!

16:55

And then it's all the right turns, right turns, right turns.

16:58

(screams)

16:59

Right here is where they take your picture.

17:01

(screams)

17:02

This is wild.

17:03

14 year old me would be so proud right now.

17:06

- This is it.

17:07

Space Mountain has been revealed.

17:09

- And thanks to Harrison's mapping technology, this means you can now actually see what it would look

17:14

like to write Space Mountain with all the lights on in CrunchLabs,

17:17
even if at this size you'd be way below the ride's 5 ft minimum height requirement.
17:22
And all of this is why I love being an engineer.
17:25
- You get to shape the future, potentially save a bunch of lives and accomplish even the most
17:31
ridiculous 30 year life goals.
17:35
If you're a teenager or an adult and you've always wanted to make and build cool stuff but just
17:39
haven't figured out that first step, this is it.
17:42
It's called the CrunchLabs Hack Pack, and it's basically a series of really fun programmable
17:46
robots that get delivered right to your door where we build it together and learn step by step the
17:51
kinds of engineering skills that go into making the builds on my YouTube channel.
17:55
And they all work with no programming required, but since my goal is to take you from wherever
17:59
you're at and level you up, you can easily hack the microcontroller brains of any of these robots
18:04
in a bunch of ways to completely level up the functionality.
18:07
There's also a community where you can share your builds or post questions, as well as an AI chatbot
18:12
named Mark Robot that will check your code for you and help you implement your most creative ideas.
18:17
On top of all that, each box has a chance to contain a platinum diploma.
18:21
If your box has it, congratulations, because college is now free for you or a loved one you
18:25
want to transfer it to, plus you get to come out here and brainstorm one of your own ideas with me
18:29
and my team for a day.

18:30

So if you want to enhance or even just take the first step of unlocking the really fun and

18:35

rewarding hobby of making stuff, just go to CrunchLabs.com or use the link in the video

18:39

description, where to say thank you, we're giving away that free box as an early subscriber special.

18:44

- Thanks for watching.